

Assignment 1 Statistical Regression

Instructions:

- Submit your answers to the task below as a clean and well-structured report, not exceeding four (4) pages. The four pages include both the cover page and conclusion.

Note: If your submission exceeds four pages, only the first four pages will be graded.

- Submit a separate R script file containing your implementation code. The tutors will verify your results by running your R script.
- The written report must be in your own words and should not include any code within the report.

Important: Failure to follow these instructions will result in heavy penalties.

Tasks

Consider the dataset `mtcars` (Motor Trend Car Road Tests) in R. This dataset contains data from 1974 Motor Trend US magazine, about fuel consumption and 10 aspects of automobile design and performance for 32 automobiles.

The dataset contains the following variables:

<code>mpg</code>	Miles/(US) gallon
<code>cyl</code>	Number of cylinders
<code>disp</code>	Displacement (cubic inches)
<code>hp</code>	Gross horsepower
<code>drat</code>	Rear axle ratio
<code>wt</code>	Weight (1000 lbs)
<code>qsec</code>	Second to ride 1/4 mile time
<code>vs</code>	Engine (0 = V-shaped, 1 = straight)
<code>am</code>	Transmission (0 = automatic, 1 = manual)
<code>gear</code>	Number of forward gears
<code>carb</code>	Number of carburetors

1. Compute the correlations between each pair of variables and plot each variable against the others. Using correlations and plots, comment on the relations that you find more interesting. Consider in particular the correlation between `mpg` and the other variables and comment.
2. Assuming `mpg` as response variable, consider three simple linear regressions where the covariates are, respectively, `disp`, `drat` and `wt`. Estimate the regression coefficients and, for each of them, check if it is possible to reject the hypothesis that the coefficient is equal to 0. If the coefficient is significant, i.e. non zero, then comment on the effect of the corresponding covariate on `mpg`.

Note: If you use R, it would be useful to find the command which computes the confidence intervals for the regression coefficients.

3. Plot the residuals for all three regressions and comment.
4. Using the three regressions, predict `mpg` when considering, respectively, `disp`=290, `drat`=3.25 and `wt`=2.950.
5. Using the formulas given in class, compute $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\sigma}^2$ for each of the three regression models.
6. Compute and report the 95% confidence intervals for both β_0 and β_1 in each regression model. Interpret each interval in the context of the data: what do the bounds suggest about the uncertainty of the intercept and slope estimates?
7. Examine and interpret the fitted regression model and residual plots. Discuss:
 - (a) Whether the linear model appears appropriate for each covariate.
 - (b) Any patterns, non-linearity, or outliers observed in the residuals.
 - (c) What these plots indicate about the model assumptions.

End